

HIGH PERFORMANCE RESEARCH COMPUTING

Using a Passenger App for Displaying GPU
Node Configuration and Resource Availability

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Global Open OnDemand Conference

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Selecting GPUs for Interactive Apps

Number of hours (max 168)

1

Number of cores (max 96)

30

Total GB memory (max 488)

52

Node type

CPU only

CPU only

H100 GPU

A30 GPU

this field is optional.

Questions when Selecting GPUs for Interactive Apps

CONFIGURATION

- What are the current GPUs configurations on the cluster
 - how many of each type of GPU are installed per node
 - how many nodes of each GPU configuration are installed

AVAILABILITY

- How many GPUs, CPUs and GB of memory per node are currently available for interactive jobs

Current GPU node Configurations on HPRC Clusters

ACES*+

gpu:h100:2
gpu:h100:4
gpu:h100:8
gpu:a30:2
gpu:pvc:2
gpu:pvc:4
gpu:pvc:8

FASTER*+

gpu:a100:4
gpu:a100:5
gpu:a100:6
gpu:a100:8
gpu:a100:16
gpu:t4:4
gpu:t4:8
gpu:a40:2
gpu:a40:4
gpu:a30:2
gpu:a10:2
gpu:a10:4

Grace

gpu:a100:2
gpu:a40:3
gpu:rtx:2
gpu:t4:4

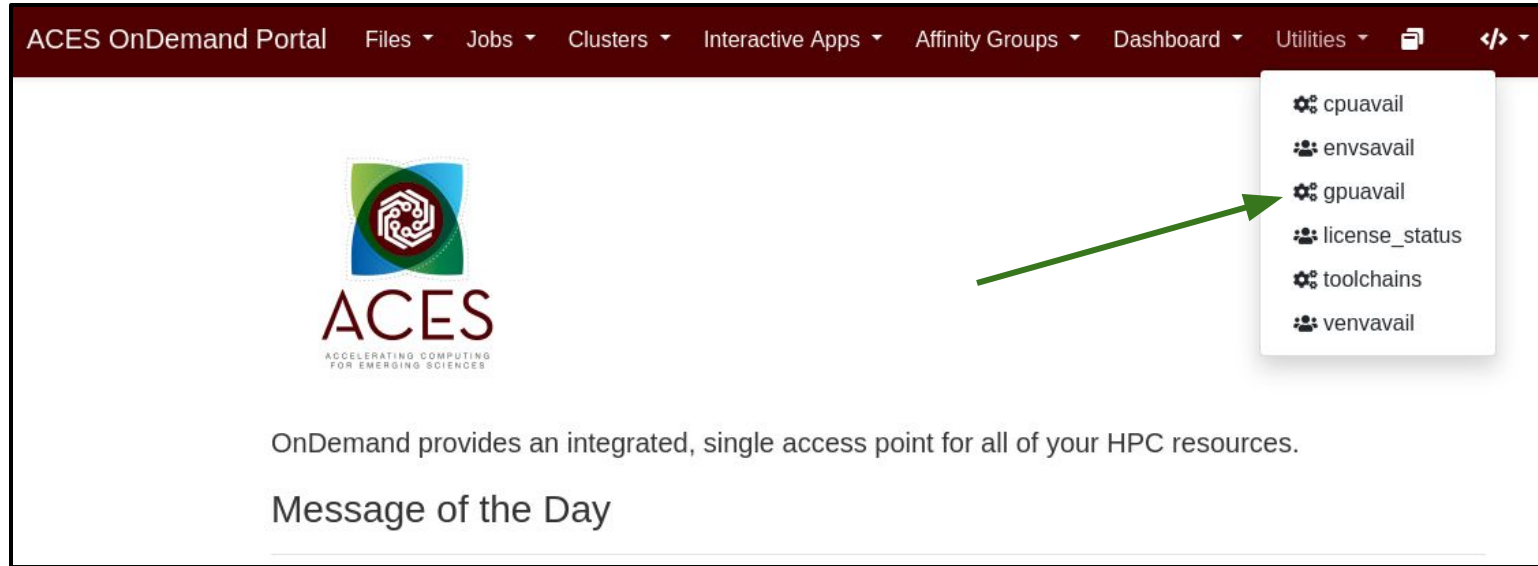
Launch+

gpu:a30:2

* Hardware composable clusters

+ available through ACCESS <https://access-ci.org>

gpuavail Accessible as a Utility



The screenshot shows the ACES OnDemand Portal interface. The top navigation bar includes links for Files, Jobs, Clusters, Interactive Apps, Affinity Groups, Dashboard, and Utilities. The Utilities dropdown menu is open, displaying a list of utilities: cpuavail, envsavail, gpuavail, license_status, toolchains, and venvavail. A green arrow points to the 'gpuavail' option. The main content area features the ACES logo, which consists of a stylized circuit icon above the text 'ACES' and the tagline 'ACCELERATING COMPUTING FOR EMERGING SCIENCES'. Below the logo, a message states: 'OnDemand provides an integrated, single access point for all of your HPC resources.' and 'Message of the Day'.

ACES OnDemand Portal Files Jobs Clusters Interactive Apps Affinity Groups Dashboard Utilities

cpuavail
envsavail
gpuavail
license_status
toolchains
venvavail

ACES
ACCELERATING COMPUTING
FOR EMERGING SCIENCES

OnDemand provides an integrated, single access point for all of your HPC resources.

Message of the Day

ACES GPU Availability

This displays the CONFIGURATION of installed GPUs and the status of GPUs readily available for jobs.

GPUs and GPU nodes not in the AVAILABILITY table are busy with other jobs and will be available after jobs have completed.

Output generated: **2025-03-15 20:00:14**

```
$ gpuavail
```

CONFIGURATION		AVAILABILITY					
NODE TYPE	NODE COUNT	NODE NAME	GPU TYPE	GPU COUNT	GPU AVAIL	CPU AVAIL	GB MEM AVAIL
gpu:pvc:4	17	ac064	a30	2	1	88	456
gpu:pvc:2	11	ac013	pvc	4	4	96	488
gpu:pvc:8	4	ac024	pvc	8	8	96	488
gpu:h100:4	3	ac100	pvc	2	2	96	488
gpu:h100:8	2	ac101	pvc	4	4	96	488
gpu:h100:2	1	ac082	pvc	2	2	96	488
gpu:a30:2	1	ac085	pvc	4	4	96	488
		ac099	pvc	4	4	96	488
		ac103	pvc	2	2	96	488

ACES GPU Availability

This displays the CONFIGURATION of installed GPUs and the status of GPUs readily available for jobs.

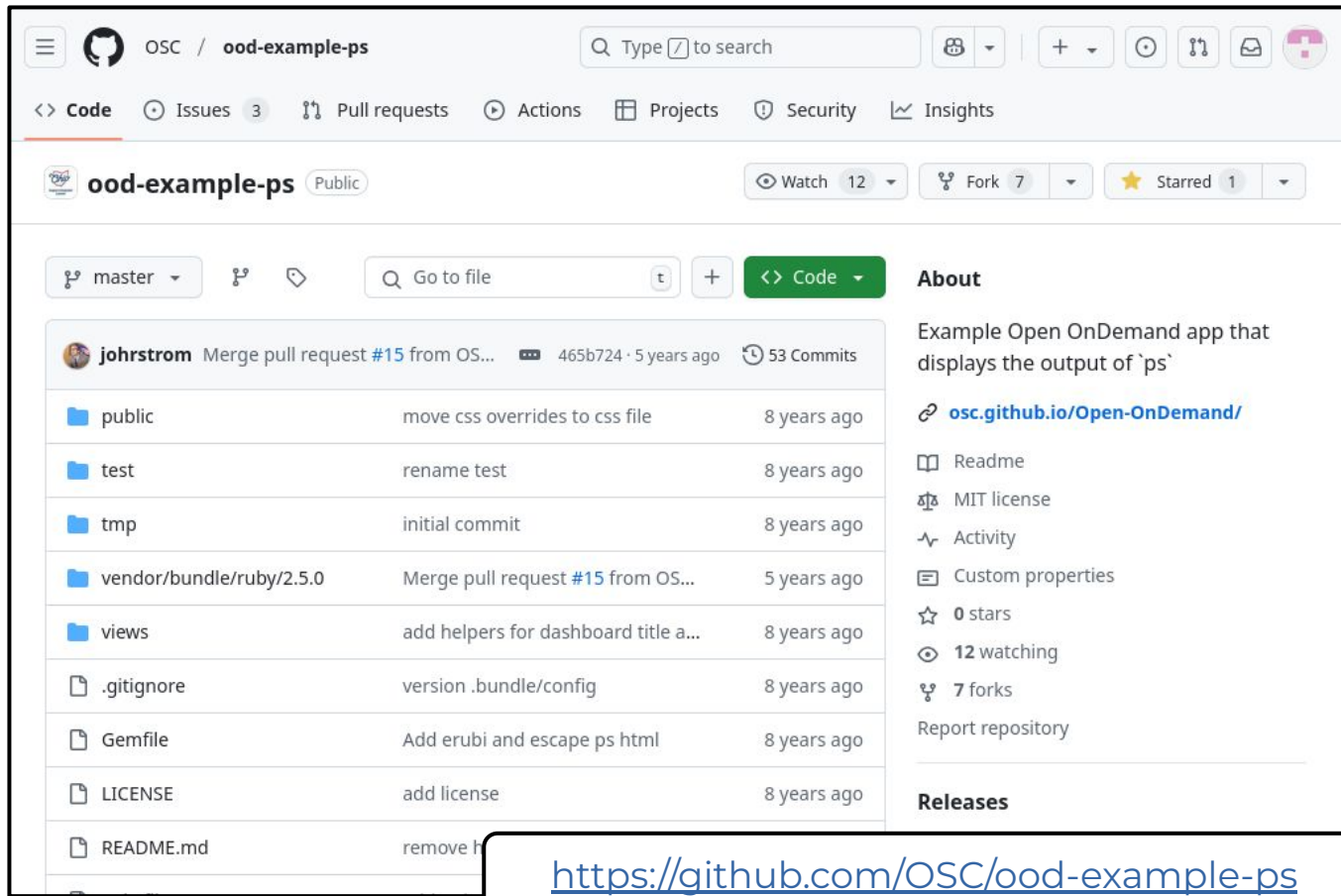
GPUs and GPU nodes not in the AVAILABILITY table are busy with other jobs and will be available after jobs have completed.

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CONFIGURATION		AVAILABILITY					
NODE TYPE	NODE COUNT	NODE NAME	GPU TYPE	GPU COUNT	GPU AVAIL	CPU AVAIL	GB MEM AVAIL
gpu:pvc:4	17	ac064	a30	2	1	88	456
gpu:pvc:2	11	ac013	pvc	4	4	96	488
gpu:pvc:8	4	ac024	pvc	8	8	96	488
gpu:h100:4	3	ac100	pvc	2	2	96	488
gpu:h100:8	2	ac101	pvc	4	4	96	488
gpu:h100:2	1	ac082	pvc	2	2	96	488
gpu:a30:2	1	ac085	pvc	4	4	96	488
		ac099	pvc	4	4	96	488
		ac103	pvc	2	2	96	488

gpuavail utilizes code from the ood-example-ps github repo which is an app that displays the output of the **ps** command

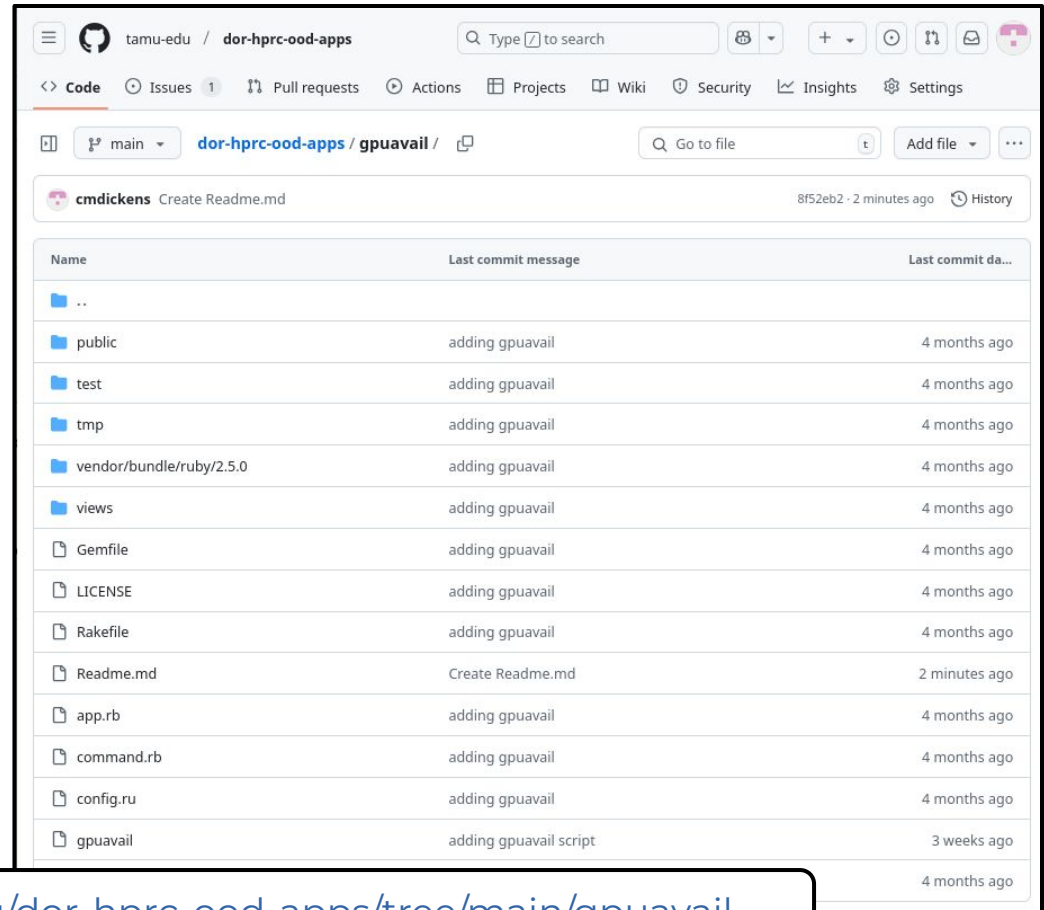


The screenshot shows the GitHub repository page for `ood-example-ps` by the organization `OSC`. The repository is public and has 12 watchers, 7 forks, and 1 star. The main content area displays a list of files and folders with their commit history. The files include `public`, `test`, `tmp`, `vendor/bundle/ruby/2.5.0`, `views`, `.gitignore`, `Gemfile`, `LICENSE`, and `README.md`. The commit history for each file is shown, with the most recent commit being a merge pull request #15 from OSC, dated 5 years ago. The right sidebar contains the 'About' section, which describes the repository as an 'Example Open OnDemand app that displays the output of `ps`' and provides a link to `osc.github.io/Open-OnDemand/`. It also lists the repository's metadata, including the README, MIT license, activity, custom properties, 0 stars, 12 watchers, 7 forks, and a report repository link. The 'Releases' section is also visible at the bottom of the sidebar.

File	Commit Message	Commit Date
public	move css overrides to css file	8 years ago
test	rename test	8 years ago
tmp	initial commit	8 years ago
vendor/bundle/ruby/2.5.0	Merge pull request #15 from OSC...	5 years ago
views	add helpers for dashboard title a...	8 years ago
.gitignore	version .bundle/config	8 years ago
Gemfile	Add erubi and escape ps html	8 years ago
LICENSE	add license	8 years ago
README.md	remove h	8 years ago

<https://github.com/OSC/ood-example-ps>

All necessary files including the gpuavail script and OOD Passenger App files are available in the github.com tamu-edu repository



The screenshot shows the GitHub interface for the repository `tamu-edu / dor-hprc-ood-apps`. The main branch is selected, and the file tree for the `gpuavail` directory is displayed. The table below summarizes the files and their commit history.

Name	Last commit message	Last commit da...
..		
public	adding gpuavail	4 months ago
test	adding gpuavail	4 months ago
tmp	adding gpuavail	4 months ago
vendor/bundle/ruby/2.5.0	adding gpuavail	4 months ago
views	adding gpuavail	4 months ago
Gemfile	adding gpuavail	4 months ago
LICENSE	adding gpuavail	4 months ago
Rakefile	adding gpuavail	4 months ago
Readme.md	Create Readme.md	2 minutes ago
app.rb	adding gpuavail	4 months ago
command.rb	adding gpuavail	4 months ago
config.ru	adding gpuavail	4 months ago
gpuavail	adding gpuavail script	3 weeks ago

<https://github.com/tamu-edu/dor-hprc-ood-apps/tree/main/gpuavail>

gpuavail OOD App Installation Requirements

- The gpuavail script reads the output from the scontrol show node command which requires the slurm.conf to be configured using the Gres parameter
- Slurm configuration file slurm.conf must use Gres configuration parameter

NodeName=ac[049,055,096]

CPU=96 Boards=1

SocketsPerBoard=2

CoresPerSocket=48

ThreadsPerCore=1

RealMemory=500000

Gres=gpu:h100:4

...

gpuavail script also works as a
command line utility

```
[userid@aces ~]$ gpuavail
```

```

      CONFIGURATION
      NODE          NODE
      TYPE          COUNT
      -----
      gpu:pvc:4      17
      gpu:pvc:2      11
      gpu:pvc:8       4
      gpu:h100:4      3
      gpu:h100:8      2
      gpu:a30:2       1
      gpu:h100:2      1

```

```

      AVAILABILITY
      NODE          GPU   GPU   GPUs  CPUs  GB MEM
      NAME          TYPE  COUNT AVAIL  AVAIL  AVAIL
      -----
      ac064          a30   2     1     88    456
      ac082          pvc   2     2     96    488
      ac101          pvc   4     4     96    488
      ac086          pvc   2     2     96    488
      ac097          pvc   2     2     96    488
      ac039          pvc   4     4     96    488
      ac081          pvc   4     4     96    488
      ac089          pvc   4     4     96    488
      ac011          pvc   4     4     96    488
      ac023          pvc   4     4     96    488

```



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gpuavail app github repo

<https://github.com/tamu-edu/dor-hprc-ood-apps/tree/main/gpuavail>

